

DBMS WEEKLY TASK 2

Identifying Data Requirements for an Online Retail System

Primary Key

A Primary Key (PK) is a unique identifier that distinguishes each record in a database table. It prevents duplicate records and ensures that every row can be uniquely identified. A primary key cannot contain NULL values and is used to establish relationships between different tables.

Example:

Every customer is assigned a unique Customer_ID (e.g., CUS001). Even if two customers have the same name, their Customer_ID will always be different.

Main Data Entities

1. Customer

Stores customer information.

Attributes:

Customer_ID (Primary Key)

Customer_Name

Email

Phone_Number

Password

Address

2. Product

Stores product details.

Attributes:

Product_ID (Primary Key)

Product_Name

Description

Price

Stock_Quantity

Category_ID

Supplier_ID

3. Category

Groups products into categories.

Attributes:

Category_ID (Primary Key)

Category_Name

Description

4. Supplier

Stores supplier information.

Attributes:

Supplier_ID (Primary Key)

Supplier_Name

Contact_Number

Email

Address

5. Order

Stores customer purchase details.

Attributes:

Order_ID (Primary Key)

Customer_ID

Order_Date

Order_Status

Total_Amount

6. Payment

Stores payment information.

Attributes:

Payment_ID (Primary Key)

Order_ID

Payment_Method

Payment_Amount

Payment_Status

7. Shipment

Tracks order delivery.

Attributes:

Shipment_ID (Primary Key)

Order_ID

Tracking_Number

Shipment_Status

Delivery_Date

8. Review

Stores customer feedback.

Attributes:

Review_ID (Primary Key)

Customer_ID

Product_ID

Rating

Review_Comment

Additional Data Requirements

Search History

Search_ID

Customer_ID

Conclusion:

The Online Retail System database consists of entities such as Customer, Product, Category, Supplier, Order, Payment, Shipment, and Review. Each entity has a unique Primary Key to identify records and maintain data integrity. Additional features such as Search History, Coupons, Return & Refund, Product Filters, and Sorting Options improve the system's functionality and provide a better shopping experience. This database structure supports efficient storage, retrieval, and management of online retail data.